

(12) **United States Patent**
Hingorani et al.

(10) **Patent No.:** **US 12,405,023 B2**
(45) **Date of Patent:** **Sep. 2, 2025**

(54) **HVAC SYSTEM WITH INDOOR AIR QUALITY MONITORING AND MITIGATION**

USPC 454/229
See application file for complete search history.

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 640 days.

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(21) Appl. No.: **17/741,968**

(22) Filed: **May 11, 2022**

(65) **Prior Publication Data**

US 2023/0366581 A1 Nov. 16, 2023

(51) **Int. Cl.**

F24F 11/74 (2018.01)
F24F 11/00 (2018.01)
F24F 11/63 (2018.01)
F24F 11/0/66 (2018.01)
F24F 11/0/70 (2018.01)

(52) **U.S. Cl.**

CPC **F24F 11/74** (2018.01); **F24F 11/0001** (2013.01); **F24F 11/63** (2018.01); **F24F 2110/66** (2018.01); **F24F 2110/70** (2018.01)

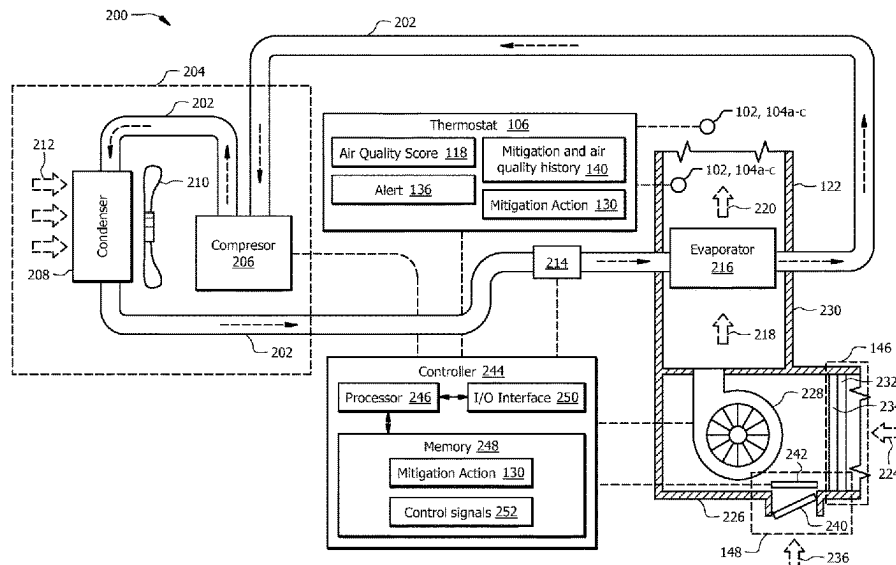
(58) **Field of Classification Search**

CPC **F24F 11/74**; **F24F 11/63**; **F24F 11/0001**; **F24F 2110/70**; **F24F 2110/66**

(57) **ABSTRACT**

An HVAC system includes one or more air quality sensors, each configured to measure an air quality and a thermostat communicatively coupled to the one or more air quality sensors. The thermostat receives indoor air quality measurements from the one or more air quality sensors. An indoor air quality score is determined based at least in part on the received indoor air quality measurements. The thermostat determines, based at least in part on the indoor air quality score, a mitigation action, wherein the mitigation action comprises one or more actions selected from the group of: (i) a filtering action comprising filtering air provided to the space using an air purification subsystem, and (ii) a ventilation action comprising ventilating the space using a ventilation subsystem. The mitigation action is executed, or implemented, by adjusting one or more components of the HVAC system.

17 Claims, 6 Drawing Sheets



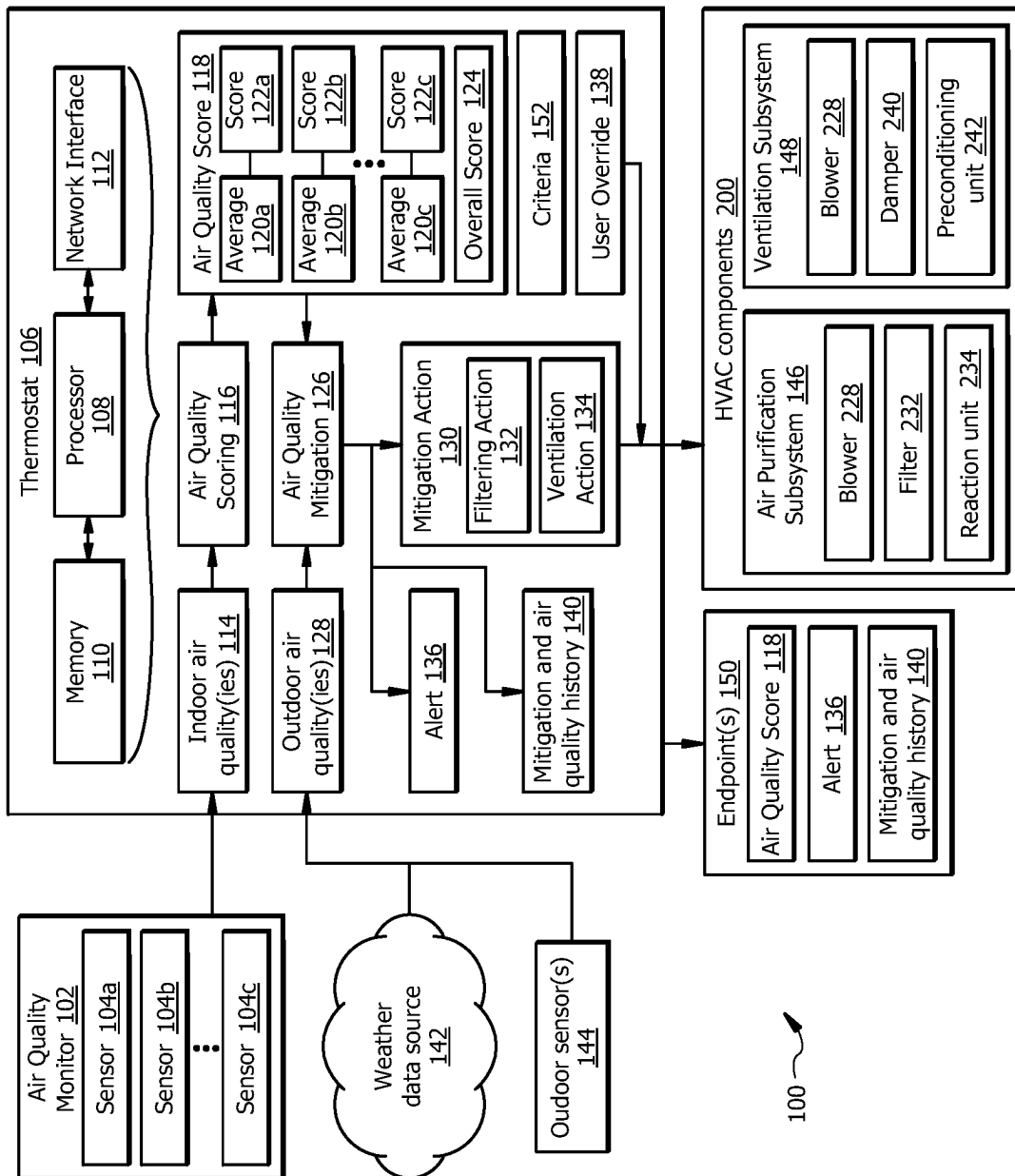


FIG. 1

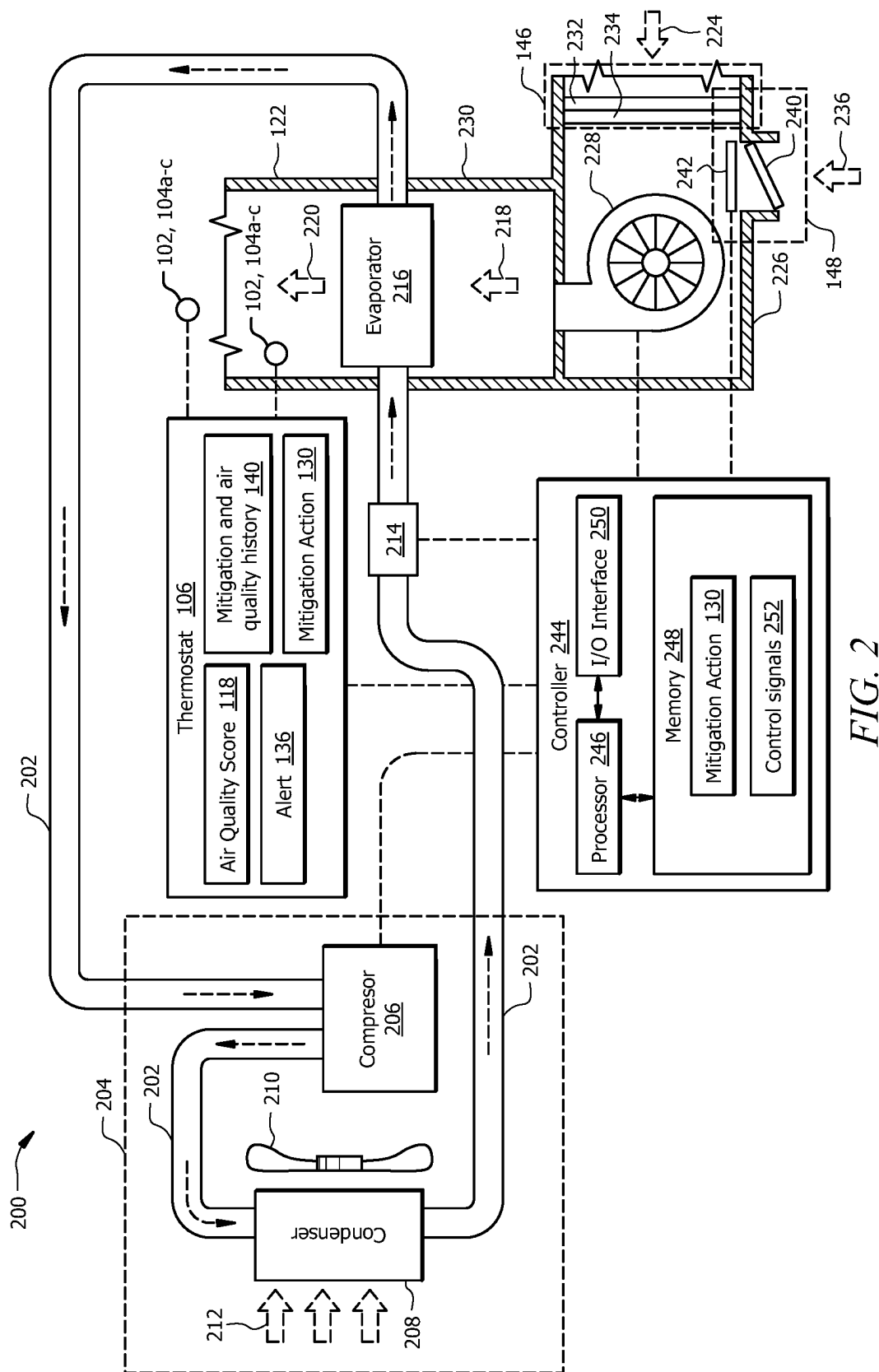


FIG. 2

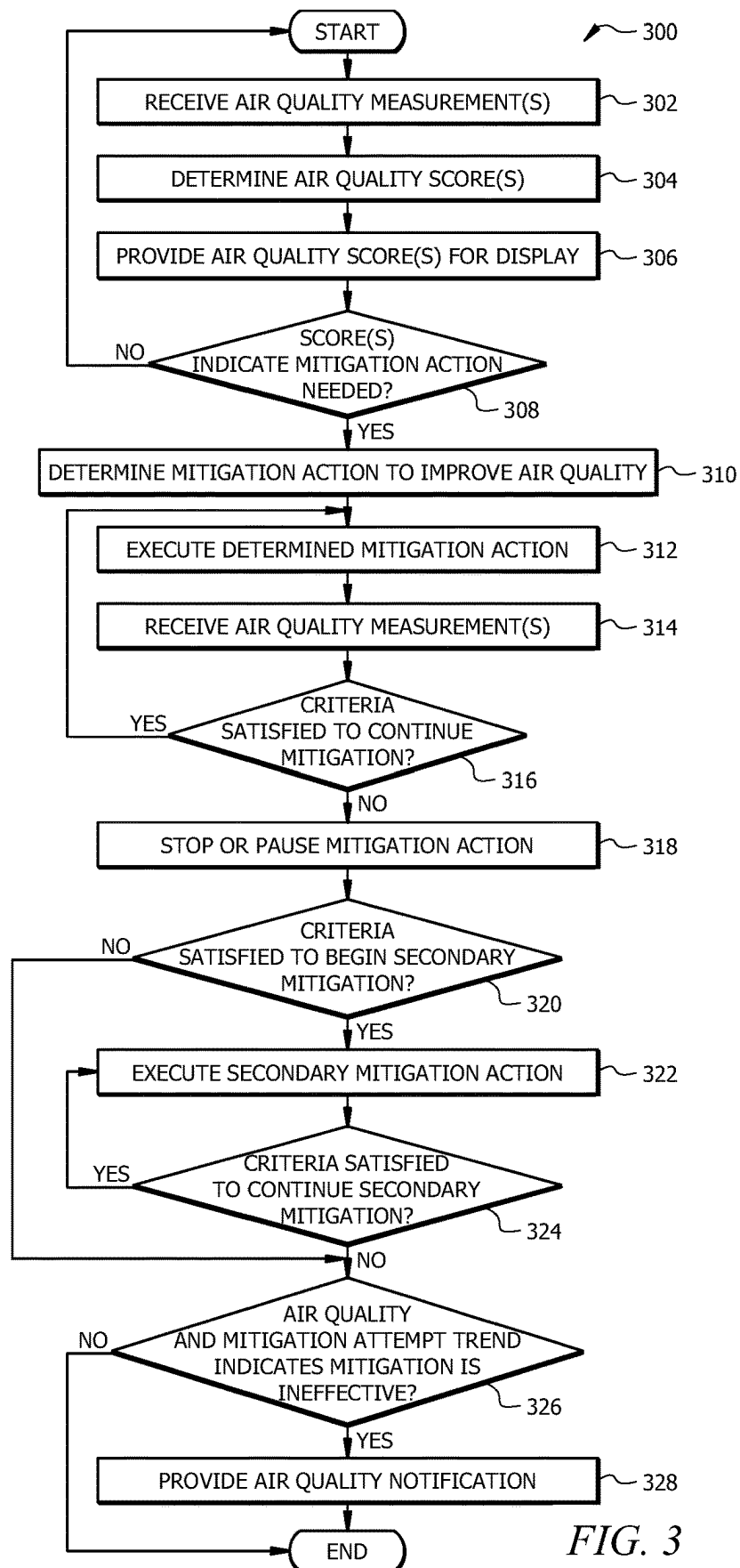


FIG. 3

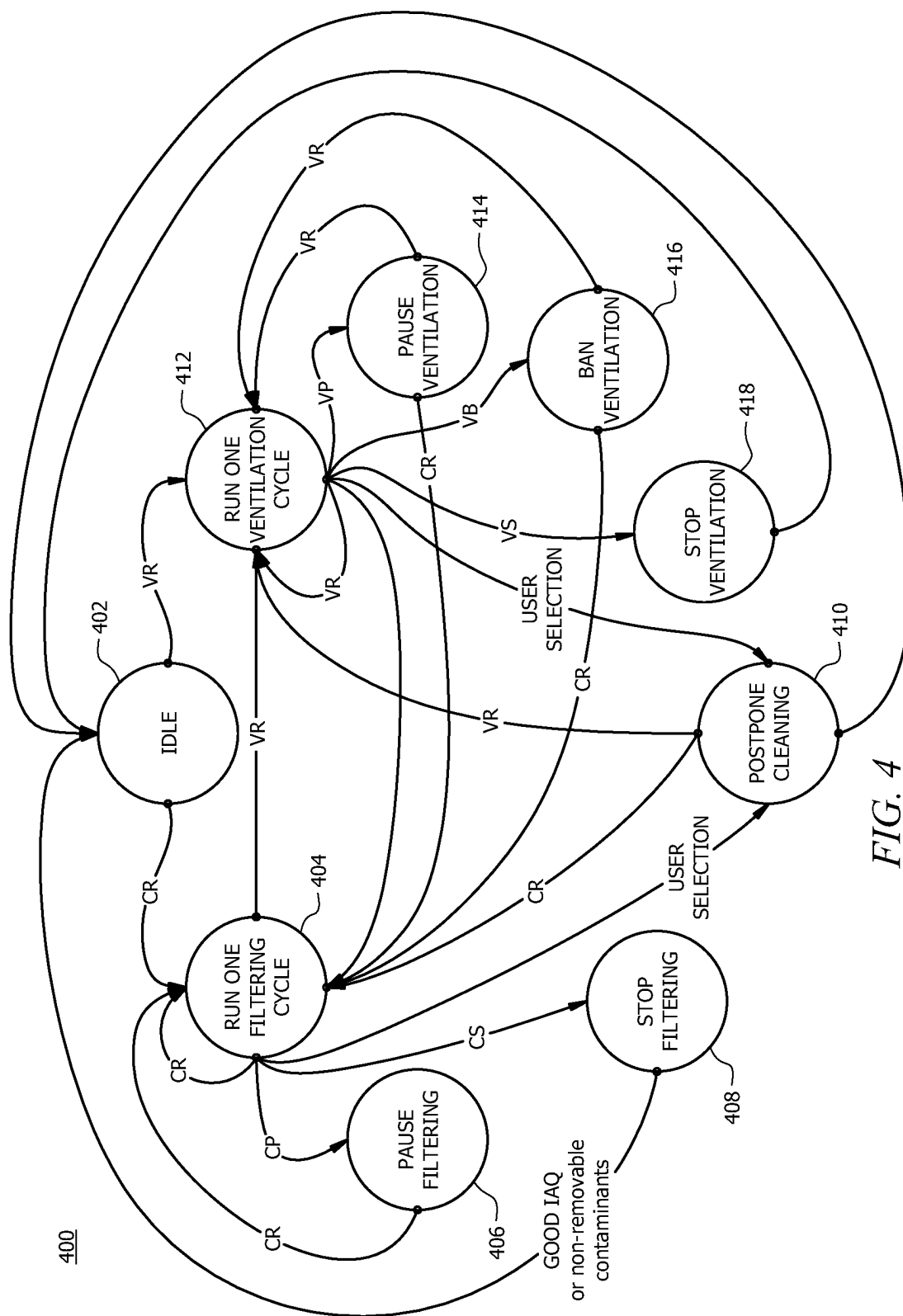


FIG. 4

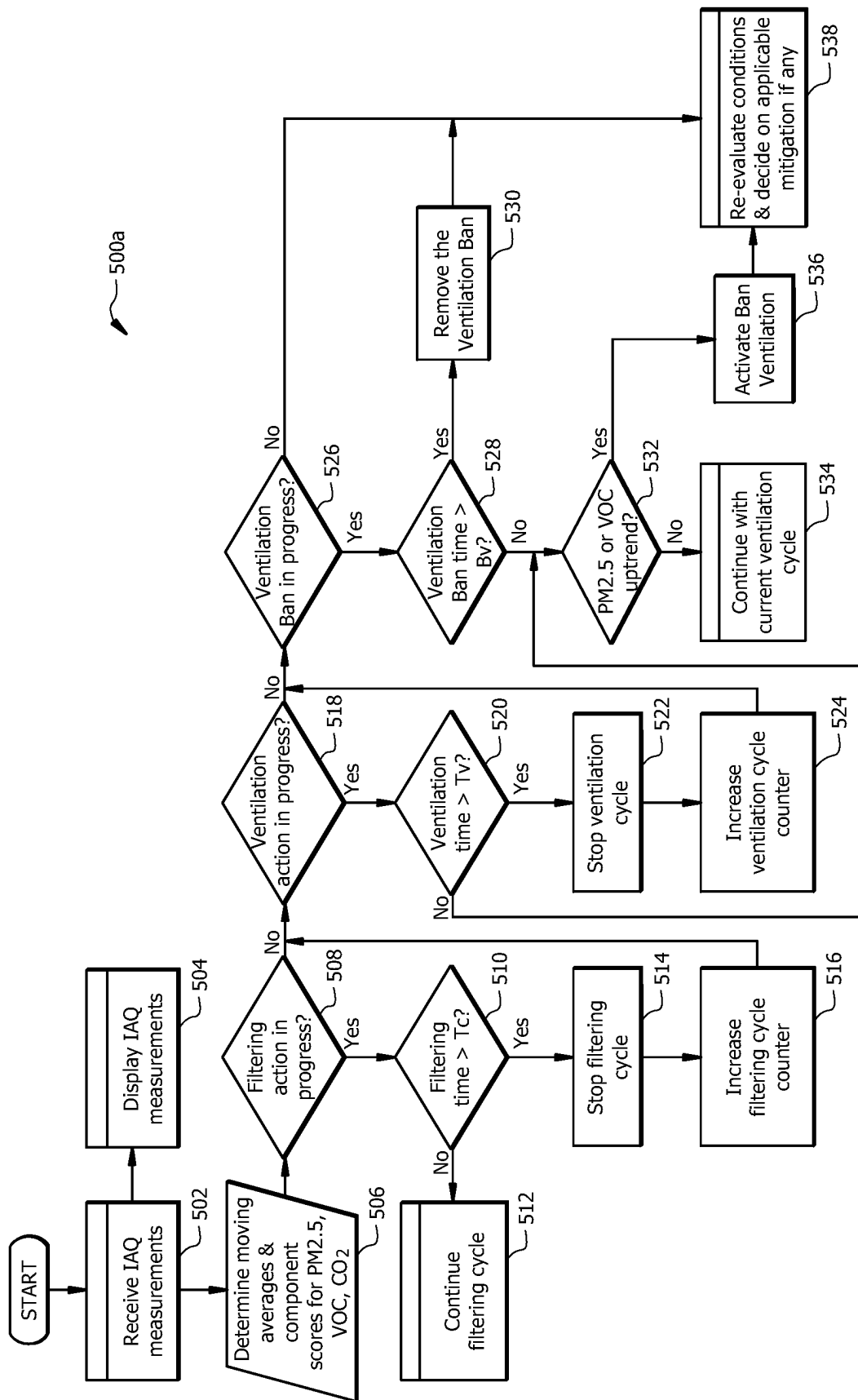
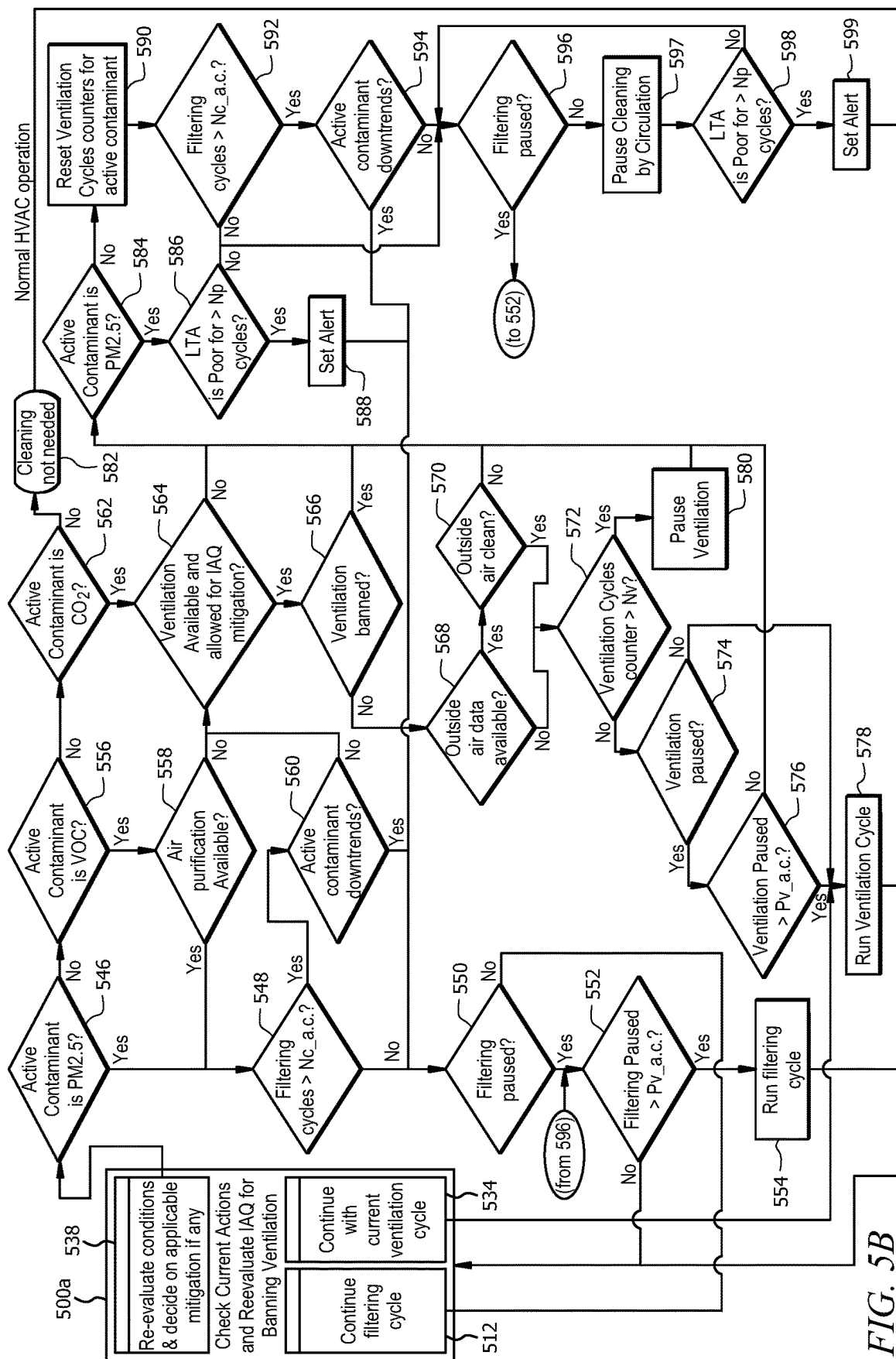


FIG. 5A



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HVAC SYSTEM WITH INDOOR AIR QUALITY MONITORING AND MITIGATION

TECHNICAL FIELD

This disclosure relates generally to heating, ventilation, and air conditioning (HVAC) systems. More particularly, in certain embodiments, this disclosure relates to an HVAC system with indoor air quality monitoring and mitigation.

BACKGROUND

Heating, ventilation, and air conditioning (HVAC) systems are used to regulate temperature within an enclosed space. Air is cooled via heat transfer with refrigerant flowing through the HVAC system and returned to the enclosed space as conditioned air.

SUMMARY OF THE DISCLOSURE

HVAC systems generally provide cooled or heated air to a space to improve comfort of occupants of the space. This disclosure recognizes that other qualities of air in the conditioned space can also be monitored, and that intelligent actions can be taken to improve indoor air quality. To achieve this and other improvements over previous technology, this disclosure provides an intelligent thermostat that receives information about indoor air quality and uses this information to provide efficient and reliable remediation of any air qualities that are outside a desired range. Mitigation actions are selected based on real-time indoor and/or outdoor air quality metrics, recent trends of individual contaminant levels, and/or previous air cleaning attempts, such that the most efficient and effective mitigations are executed to improve the air quality in a space. Altogether the improved thermostat facilitates an automated approach to maintaining air quality at desired levels, while reporting air quality information to the users via real-time visual indicators and with retrospective/historical data. For example, trends in attempted mitigations and associated air quality can be used to identify system failures or flag unhealthy air quality exposures. Certain embodiments may include none, some, or all of the above technical advantages. One or more other technical advantages may be readily apparent to one skilled in the art from the figures, descriptions, and claims included herein.

In an embodiment, an HVAC system includes one or more air quality sensors, each configured to measure an air quality and a thermostat communicatively coupled to the one or more air quality sensors. The thermostat receives indoor air quality measurements from the one or more air quality sensors. An indoor air quality score is determined based at least in part on the received indoor air quality measurements. The thermostat determines, based at least in part on the indoor air quality score, a mitigation action, wherein the mitigation action comprises one or more actions selected from the group of: (i) a filtering action comprising filtering air provided to the space using an air purification subsystem, and (ii) a ventilation action comprising ventilating the space using a ventilation subsystem. The mitigation action is executed, or implemented, by adjusting one or more components of the HVAC system.

BRIEF DESCRIPTION OF THE DRAWINGS

For a more complete understanding of the present disclosure, reference is now made to the following description, taken in conjunction with the accompanying drawings, in which:

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FIG. 1 is a block diagram of an example HVAC system configured for improved indoor air quality monitoring and mitigation;

FIG. 2 is a diagram illustrating the HVAC components of the system of FIG. 1 in more detail;

FIG. 3 is a flowchart of an example method of operating the system of FIG. 1;

FIG. 4 is a flow diagram illustrating transitions between various operating states of the HVAC system of FIG. 1 for improved air quality mitigation; and

FIGS. 5A and 5B are flowcharts illustrating other example methods of operating the system of FIG. 1.

DETAILED DESCRIPTION

Embodiments of the present disclosure and its advantages are best understood by referring to FIGS. 1 through 5B of the drawings, like numerals being used for like and corresponding parts of the various drawings.

As described above, prior to the present disclosure, there was a lack of tools for efficiently and automatically monitoring air quality and mitigating any detected air quality issues. This disclosure provides technical solutions to these and other problems by providing an intelligent thermostat or HVAC system controller that monitors indoor air quality (e.g., via sensor measurements), determines indoor air quality scores for various quality types (or corresponding contaminants such as particulate matter, volatile organic compounds, carbon dioxide, and the like), determines a mitigation action for improving any air quality scores below a threshold, and automatically executes the mitigation action. The indoor air quality continues to be monitored after and/or while the mitigation action is performed, and the mitigation action may be repeated, paused, stopped, or changed depending on how the indoor air quality is impacted by attempted mitigation actions. In some cases, the mitigation action is performed in cycles, and a reassessment is performed to determine if the same mitigation action should be repeated, mitigation should be stopped, or a different mitigation action should be performed (see, e.g., FIGS. 4, 5A and 5B).

HVAC System

FIG. 1 shows a block diagram of an example HVAC system 100 that includes at least one air quality monitor 102, a thermostat 106 configured for air quality monitoring and mitigation, HVAC components 200 (see FIG. 2) operable to provide conditioned air to a space and execute mitigation actions 130 determined by the thermostat 106, and endpoints 150 capable of presenting information determined by the thermostat 106. The HVAC system 100 is generally configured to deliver conditioned air to a space, as illustrated in greater detail with respect to FIG. 2, described below, which shows other HVAC components 200 in greater detail.

Air Quality Data Sources—Air Quality Monitor, Weather Data Source, and Outdoor Sensor(s)

The air quality monitor 102 may include one or more air quality sensors 104a-c. The example of FIG. 1 shows three air quality sensors 104a-c. However, this disclosure contemplates the HVAC system 100 including any number of air quality sensors 104a-c. Each air quality sensor 104a-c measures a property of the indoor air in the space serviced by the HVAC system 100. For instance, a first air quality sensor 104a may be a particle sensors that measures a particle concentration in the space. Examples of such sensors are particulate matter (PM) sensors that measure particle concentrations for different particle sizes (e.g., values of PM1.0, PM2.5, and the like). As another example, a

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second air quality sensor **104b** may be a gas sensors that measures total volatile organic compounds (TVOCs or simply VOCs) in the indoor space. TVOC is a measure of the level, or concentration, of total measurable organic compounds in the air. As yet another example, a third air quality sensor may be a gas sensor that measures carbon dioxide (CO₂) in the indoor space. As further non-limiting examples, air quality sensors **104a-c** may be other sensors to measure any component that may be found in air and considered a contaminant, such as carbon monoxide, ozone, radioactivity, radon, formaldehydes, nitrogen oxides, and the like.

In some cases, a sensor **104a-c** may be located within a duct of the HVAC system **100**, as shown in the example of FIG. 2 in which a sensor **104a-c** is located within duct **222**. In such cases, measurements of indoor air quality **114** may only be obtained while the blower **228** is running to provide a flow of air through the duct (e.g., airflow **220** of FIG. 2). Measurements of indoor air quality **114** may not be provided to the thermostat **106** when the blower **228** is not running or has not run for a sufficiently long period of time. If a measurement of indoor air quality **114** is needed but the blower **228** is not running, the thermostat **106** may temporarily cause the blower **228** to turn on or run so that a measurement of indoor air quality **114** can be obtained from an in-duct sensor **104a-c**. For example, the thermostat **106** (described in greater detail below) may wait until a minimum air flow (see CFMm of TABLE 1 below) is established before using air qualities **114** measured from an in-duct sensor **104a-c**. The thermostat **106** may also or alternatively wait until the blower **228** has operated for a minimum runtime (see Tw of TABLE 1 below) before using air qualities **114** measured from an in-duct sensor **104a-c** and/or delay collection of air qualities **114** until the blower **228** has operated for a minimum ramp-up time (see Tr of TABLE 1 below). The thermostat **106** may operate according to a maximum time interval (see Tm of TABLE 1 below) between receiving measures of air qualities **114** from an in-duct sensor **104a-c**. If a measurement of an air quality **114** is not received within that interval, the blower **228** is powered on to take in-duct measurement of the air quality **114**.

As described further below, in some cases, outdoor air qualities **128** may be used to help determine an appropriate mitigation action **130** for improving indoor air qualities **114**. In some cases, outdoor air properties **128** may be provided by or determined from information provided by a weather data source **142**. For example, a weather data source **142** may provide an indication of an outdoor air quality **128** in the form of an outdoor air quality index. In some cases, one or more outdoor sensors **144** may be used to determined outdoor air qualities. The outdoor sensors **144** may be the same as or similar to sensors **104a-c** described above.

Thermostat

The thermostat **106** is in communication (e.g., via wireless and/or wired communication) with the air quality monitor **102** to receive indoor air qualities **114** and with HVAC components **200** to implement or execute a mitigation action **130** to remediate any detected air quality issues. Turning briefly to FIG. 2, the thermostat **106** may be in communication with a controller **244** of the HVAC system **100**. In some embodiments, all or a portion of the operations of the controller **244** may be performed by the thermostat **106**. Referring to both FIGS. 1 and 2, the thermostat **106** receives measurements of indoor air quality **114** from the one or more air quality sensors **104a-c**. The air qualities **114** may, for example, be the concentrations of different contaminants

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that are detected by the air quality sensors **104a-c** (e.g., PM, TVOC, CO₂, and/or the like).

The thermostat **106** includes a processor **108**, memory **110**, and network interface **112**. The processor **108** includes one or more processors. The processor **108** is any electronic circuitry including, but not limited to, state machines, one or more central processing unit (CPU) chips, logic units, cores (e.g., a multi-core processor), field-programmable gate array (FPGAs), application specific integrated circuits (ASICs), or digital signal processors (DSPs). The processor **108** may be a programmable logic device, a microcontroller, a microprocessor, or any suitable combination of the preceding. The processor **108** is communicatively coupled to and in signal communication with the memory **110** and network interface **112**. The one or more processors are configured to process data and may be implemented in hardware and/or software. For example, the processor **108** may be 8-bit, 16-bit, 32-bit, 64-bit or of any other suitable architecture.

The memory **110** is operable to store any data, instructions, logic, rules, or code operable to execute the functions of the thermostat **106**, including, but not limited to, the items illustrated in FIG. 1. The memory **110** includes one or more disks, tape drives, or solid-state drives, and may be used as an over-flow data storage device, to store programs when such programs are selected for execution, and to store instructions and data that are read during program execution. The memory **110** may be volatile or non-volatile and may comprise read-only memory (ROM), random-access memory (RAM), ternary content-addressable memory (TCAM), dynamic random-access memory (DRAM), and static random-access memory (SRAM).

The network interface **112** is configured to enable wired and/or wireless communications. The network interface **112** is configured to communicate data between the thermostat **106** and other network devices, systems, or domain(s), such as the air quality monitor **102**, weather data source **142**, outdoor sensor(s) **144**, HVAC components **200**, and endpoints **150**. The network interface **112** is an electronic circuit that is configured to enable communications between devices. For example, the network interface **112** may include one or more serial ports (e.g., USB ports or the like) and/or parallel ports (e.g., any type of multi-pin port) for facilitating this communication. As a further example, the network interface **112** may include a WIFI interface, a local area network (LAN) interface, a wide area network (WAN) interface, a modem, a switch, or a router. The processor **108** is configured to send and receive data using the network interface **112**. The network interface **112** may be configured to use any suitable type of communication protocol.

Air Quality Scoring

Referring again to FIG. 1, the thermostat **106** receives indoor air qualities **114** (e.g., levels of PM, TVOC, CO₂, and/or the like) from the air quality monitor **102**. The sensors **104a-c** of the air quality monitor **102** may measure air qualities **114** at a predefined rate (see IAQSamplingRate of TABLE 1 below). Measurements of air qualities **114** from the sensors **104a-c** may be transmitted at a predefined frequency (see Tt of TABLE 1 below). The thermostat **106** may define a maximum interruption time (see Ti of TABLE 1 below) after which measurements of air qualities **114** are deemed unavailable if not received from sensors **104a-c**.

After receiving measurements of air qualities **114**, the thermostat **106** performs air quality scoring **116** to determine an air quality score **118** based on the indoor air qualities **114**. For example, the thermostat **106** may determine an average value **120a-c** for each indoor air quality **114** that is measured. The average value **120a-c** may be a moving average

of sensor measurements of the air qualities **114** over a period of time. Averaging may, for example, remove or decrease the influence of erratic sensor readings. Each average value **120a-c** is converted to a corresponding score **122a-c** (e.g., of good, fair, or poor). For example, each average value **120a-c** may be compared to a corresponding threshold value (see, e.g., Good_X and Poor_X of TABLE 1 below, where X identifies the contaminant) for the air quality based on adjustable or predefined thresholds for each contaminant or corresponding air quality **114**. An overall score **124** may be determined, such that a concise, single indication may be reported (e.g., via an endpoint device **150**) to communicate air quality to occupants of the space or others. For example, the overall score **124** may be reported to endpoint devices **150** that are a user-interface of the thermostat **106**, a building automation system, a building log system, or the like. The overall score **124** may be the lowest of the scores **122a-c**. Individual scores **122a-c** may similarly be available for review via an endpoint device **150**. If any air qualities **114** (e.g., from one or more of the sensors **104a-c**) are not available for any reason, the thermostat **106** proceeds with air quality scoring **116** using the available measurements of air qualities **114**.

Example parameters and example values of these parameters that may be used to tune air quality scoring **116** are shown in TABLE 1 below. These parameters may be adjusted, for example, based on field trial performance or initial installation feedback to improve the usefulness of air quality scoring **116** for initiating automatic mitigations, as described further below. As a non-limiting example, the following three parameters may be used to determine a score **122a-c** for each contaminant or air quality **114**: (1) Good_X is an upper boundary for acceptable levels of contaminants, (2) Poor_X is a lower boundary for poor levels of contaminants, and (3) Scale_X is used to adjust to baseline levels of each contaminant for a given site or location. X represents each contaminant or air quality **114** measured by sensors **104a-c**. For example, X may represent a concentration of PM, a concentration of TVOC, a concentration of CO₂, or the like. Scale_X may provide unique benefits because expected baseline values of air qualities **114** may be different based on environmental factors (e.g., proximity to dust-producing processes and/or expected exposure to certain contaminants. For instance, in various locations and indoor spaces, the baseline levels of contaminants may vary, such that a general default Good_X value may not provide the best representative value beyond which sub-optimal air quality should be reported and a mitigation action **130** should be triggered.

TABLE 1

Example parameters for scoring indoor air quality					
Parameter	Purpose	Units	Min	Max	Default
IAQSamplingRate	Sensor data sampling rate	1/min	1	60	2
Tt	Air quality monitor transmission frequency	1/min	1	5	1
Ti	max data interruption time after which data are deemed as unavailable	min	3	10	5
Tsa	time span over which sensor measurements are aggregated	min	1	30	5

TABLE 1-continued

Example parameters for scoring indoor air quality					
Parameter	Purpose	Units	Min	Max	Default
Tla	for the short-term averages time span over which sensor measurements are aggregated for the long-term averages	day	1	30	7
Tm	max time interval between polling data from in-duct sensors	min	3	45	15
CFMm	min airflow required for sensor reading for in-duct sensors	CFM	CFM min	CFM max	CFM min
Tw	min blower runtime to gather data from in-duct sensors	min	Tsa	Tsa × 2	Tsa + 1
Tr	delay before collecting measurements to allow for blower ramp-up	sec	0	60	30
Good_PM2.5	good PM2.5 level upper boundary	µg/m ³	0	15	12
Poor_PM2.5	poor PM2.5 level lower boundary	µg/m ³	36	100	36
Good_CO2	good CO ₂ level upper boundary	ppm	400	1000	800
Poor_CO2	poor CO ₂ level lower boundary	ppm	2000	7000	2000
Good_VOC	good VOC level upper boundary	ppb	0	500	220
Poor_VOC	poor VOC level lower boundary	ppb	660	2200	1000
Scale_PM2.5	PM2.5 boundary good level scaling factor	#	0.5	3	1
Scale_VOC	VOC boundary scaling factor	#	0.5	3	1
Scale_CO2	CO ₂ boundary scaling factor	#	0.5	3	1

To achieve reliable scaling of Good_X and Poor_X, Scale_X values are selected to adjust for these location-based considerations. For example, a technician installing the HVAC system **100** in a given location may adjust values of Scale_X for each contaminant or air quality **114**. For instance, in a location with relatively clean air in terms of volatile organic compounds (Air Cleanness Level of “High”), Scale_X values for air qualities **114** corresponding to particulate matter (scale_PM), TVOC (scale_VOC), and carbon dioxide (scale_CO2) may be set to scale_PM=1, scale_VOC=1, and scale_CO2=1, respectively. In a location with moderately clean air in terms of volatile organic compounds and particulate matter (Air Cleanness Level of “Moderate”), Scale_X values may be set to scale_PM=1.1, scale_VOC=1.59, and scale_CO2=1. In a location with relatively poor air quality (Air Cleanness Level of “Basic”), Scale_X values may be set to scale_PM=1.3, scale_VOC=2.25, and scale_CO2=1. Example values of Good_X, Poor_X, and Scale_X for different example air qualities **114** or corresponding contaminants are illustrated in TABLE 1 above.

Scores **122a-c** may be determined based on averages **120a-c**. Averages **120a-c** may be short-term average (STAs) of the different air qualities **114**. The averages **120a-c** may be a moving average of all data points collected during a minimum time period (see, e.g., Tsa of TABLE 1 above). To determine the score **122a-c** from each STA corresponding to averages **120a-c**, each average **120a-c** may be compared to a scaled threshold value. For instance, the score **122a-c** for a given air quality may be “Good” the average **120a-c** for

that air quality **114** (STA_x) is less than or equal to the product of Scale_xxGood_x. A “Fair” score **122a-c** may be determined if Scale_xxGood_x < STA_x < Scale_xxPoor_x. A “Poor” score **122a-c** may be determined if STA_x ≥ Scale_xxPoor_x. A score **122a-c** of “unavailable” if STA_x = NaN (not a number), for example, because of a sensor reading error or other error. The overall score **124** is determined to summarize the individual scores **122a-c** for the measured air qualities **114**. As an example, an overall score **124** of “Good” may be determined if all of the individual scores **122a-c** are “Good”. The overall score **124** may be “Poor” if any of the individual scores **122a-c** is “Poor”. Otherwise, the overall score **124** may be “Fair”. In some cases, one or more of the averages **120a-c** may be a long-term average (LTA_x) of an indoor air quality **114**. Long-term averages may be determined similarly to STAs but over a longer period of time (e.g., as defined by T1a of TABLE 1 above).

Air Quality Mitigation

The thermostat **106** then uses the air quality score **118** to perform air quality mitigation **126** by determining and executing a mitigation action **130**. The mitigation action **130** may include a filtering action **132**, which involves filtering air that is provided to the space by the HVAC system **100** using an air purification subsystem **146**, and/or a ventilation action **134**, which involves ventilating the space using a ventilation subsystem **148**, as described in greater detail below. In some cases, the mitigation action **130** is determined at least in part using one or more outdoor air qualities **128**. For example, a ventilation action **134** may be prevented if the outdoor air quality **128** is poor. TABLE 2 summarizes when a filtering action **132** and a ventilation action **134** may be determined and performed.

TABLE 2

Summary of filtering actions 132 and ventilation actions 134.		
	Filtering Action	Ventilation Action
Primary mitigation measure for removing:	1. PM 2. TVOC if the air purification subsystem 146 includes reactor unit 234.	3. CO ₂ , 4. TVOC in the absence of a reactor unit 234.
Achieved by:	air circulation with blower 228 through air filter(s) 232.	operating the ventilation subsystem 146 to ventilate the space.
Is allowed to run continuously?	Yes. E.g., mitigation may run continuously in cases of persistent PM contamination.	No. E.g., mitigation may stop after a predefined number of back-to-back cleaning cycles, but might be re-attempted in case of persistent CO ₂ /TVOC events after a preset timeout.

An example process for determining a mitigation action **130** is described briefly here. Further examples of mitigation action **130** determination and execution are described below with respect to the examples of FIGS. 3, 4, 5A, and 5B. To determine a mitigation action **130**, the air quality score **118** may be determined every period of time (e.g., each minute) based on measurements of air qualities **114**, as described above. If the score **122a-c** for a given air quality **114** or corresponding contaminant is below a threshold (e.g., below a score **122a-c** of “Good”), that air quality **114** is selected as an active contaminant for mitigation. The mitigation action **130** is determined based on one or more of the following factors: (i) the type of air quality **114** or contaminant (e.g., PM, TVOC, CO₂) needing mitigation; (ii) the available components for providing mitigation (e.g., type of air puri-

fication subsystem **146** available, type of ventilation subsystem **148** available, or the like); (iii) trends of the active contaminant over previous mitigation attempts or cycles (e.g., whether contaminant levels are increasing, decreasing, or stable); (iv) the number of preceding mitigation attempts (e.g., whether a maximum number of mitigation cycles has been reached), (v) the outdoor air quality **128** (e.g., whether the outdoor air quality **128** is sufficiently good to allow a ventilation action **134**); and (vi) a user preference, for example, provided through a user override **138** that may pause, stop, prevent, or postpone a mitigation action **130**. The mitigation action **130** is executed by operating HVAC components **200**. Example HVAC components **200** that may be operated to perform the mitigation action **130** include an air purification subsystem **146** for implementing a filtering action **132** and a ventilation subsystem **148** for implementing a ventilation action **134**. Further details of the HVAC components **200** are described below with respect to FIG. 2.

As a further example, the thermostat **106** may determine that the mitigation action **130** is the filtering action **132** by determining that criteria **152** are satisfied that are associated with one or more of a current indoor air quality **114** (e.g., a current score **122a-c** for the quality **114**), a number of previous attempts to perform the filtering action **132**, and a trend in the indoor air quality **114** over time. For example, if the scores **122a-c** indicate air quality score **118** is poor and a maximum number of attempts of the filtering action **132** have not recently been performed, the thermostat **106** may determine to perform a filtering action **132**. The filtering action **132** may be executed by performing a filtration cycle using the air purification subsystem **146**. For example, the thermostat **106** may turn on blower **228** to provide a flow of

indoor air through the filter **232** and/or reactor unit **234** (see FIG. 2 and corresponding description below for further details).

In some embodiments, after completion of the filtering action **132** for a period of time or a predefined number of cycles, the thermostat **106** may reassess whether a continued mitigation is needed or whether the same or a different mitigation action **130** should be performed. If the indoor air qualities **114** received after the cycles of filtering action **132** are complete satisfy criteria **152** (e.g., indicating the filtering action **132** is not successful) associated with one or more of a type of contaminant needing to be removed, a number of previous attempts to perform the filtering action **132**, and a trend in the indoor air quality **114** over time, the filtering action **132** may be paused. The thermostat **106** determines

whether a ventilation action **134** is allowed. For example, the thermostat **106** may determine that the ventilation action **134** is allowed by determining whether criteria **152** are satisfied associated with one or more of an outdoor air quality **128**, a number of previous attempts to perform the ventilation action **134** (if any), and a trend in the indoor air quality **114** over time. For instance, if the outdoor air quality **128** is poor, the ventilation action **134** may not be allowed. If a ventilation action **134** is determined to be needed, the ventilation action **134** may be executed by, for example, opening a damper **240** of the ventilation subsystem **148** to allow a diluting flow of air to enter the space (see FIG. 2 and corresponding description below for further details).

The thermostat **106** may provide an alert **136** if an air quality score **118** has been below a threshold (e.g., included in criteria **152**) for an extended period of time (Np in Table 2). The alert **136** may indicate that air quality has been continuously poor and/or identify a particular one or more air qualities **114** that are poor. The alert **136** may be provided for display on an endpoint **150** (described below). The thermostat **106** may also determine and store in memory **110** a mitigation and air quality history **140** that is a log of air qualities **114** and air quality score **118** over time along with a record of mitigation actions **130** performed. The mitigation and air quality history **140** may be used to identify faults of the HVAC system **100** resulting in poor indoor air quality. An alert **136** may be automatically provided if the mitigation and air quality history **140** indicates that some alteration or maintenance of the HVAC system **100** is needed to improve air quality (e.g., by changing and/or servicing the air purification subsystem **146** and/or the ventilation subsystem **148**).

The thermostat **106** may be in communication with endpoints **150** through wired and/or wireless communication. Example endpoints **150** include a display of the thermostat, a mobile device, a computer associated with a supervisor of the HVAC system **100**, and the like. The endpoints may facilitate display of any information received or determined by the thermostat **106**, such as, for example, the air quality score **118**, any alerts **136**, and the mitigation and air quality history **140**.

Example HVAC Components

FIG. 2 shows a portion of the example HVAC system **100** of FIG. 1 along with further details of HVAC components **200**. The HVAC components **200** are used to condition air for delivery to a space (e.g., all or a portion of a room, a house, an office building, a warehouse, or the like). In some embodiments, the HVAC components **200** are part of a rooftop unit (RTU) that is positioned on the roof of a building, and the conditioned air is delivered to the interior of the building. In other embodiments, portion(s) of the HVAC components **200** may be located within the building and portion(s) outside the building. The HVAC components **200** may include one or more heating elements, not shown for convenience and clarity. The HVAC components **200** may be configured as shown in FIG. 1 or in any other suitable configuration. For example, the HVAC components **200** may include additional components or may omit one or more components shown in FIG. 2.

The HVAC components **200** include a working-fluid conduit subsystem **202**, at least one condensing unit **204**, an expansion valve **214**, an evaporator **216**, the blower **228**, and a controller **244** coupled to thermostat **106**. The working-fluid conduit subsystem **202** facilitates the movement of a working fluid (e.g., a refrigerant) through a cooling cycle such that the working fluid flows as illustrated by the dashed arrows in FIG. 2. The working fluid may be any acceptable

working fluid including, but not limited to hydrofluorocarbons (e.g., R-410A) or any other suitable type of refrigerant.

The condensing unit **204** includes a compressor **206**, a condenser **208**, and a fan **210**. In some embodiments, the condensing unit **204** is an outdoor unit while other HVAC components **200** may be located indoors. In typical embodiments, the compressor **206** is a variable speed compressor that can be operated at a range of speeds. The compressor **206** is coupled to the working-fluid conduit subsystem **202** and compresses (i.e., increases the pressure of) the working fluid. The compressor **206** is in signal communication with the controller **244** using wired and/or wireless connection. The controller **244** provides commands and/or signals to control operation of the compressor **206** and/or receive signals from the compressor **206** corresponding to a status of the compressor **206**.

The condenser **208** is configured to facilitate movement of the working fluid through the working-fluid conduit subsystem **202**. The condenser **208** is generally located downstream of the compressor **206** and is configured to remove heat from the working fluid. The fan **210** is configured to move air **212** across the condenser **208**. For example, the fan **210** may be configured to blow outside air through the condenser **208** to help cool the working fluid flowing therethrough. The fan **210** may be in communication with the controller **244** (e.g., via wired and/or wireless communication) to receive control signals for turning the fan **210** on and off and/or adjusting a speed of the fan **210**. The compressed, cooled working fluid flows from the condenser **208** toward the expansion valve **214**.

The expansion valve **214** is coupled to the working-fluid conduit subsystem **202** downstream of the condenser **208** and is configured to remove pressure from the working fluid. In this way, the working fluid is delivered to the evaporator **216**. In general, the expansion valve **214** may be a valve such as an expansion valve or a flow control valve (e.g., a thermostatic expansion valve (TXV)) or any other suitable valve for removing pressure from the working fluid while, optionally, providing control of the rate of flow of the working fluid.

The evaporator **216** is generally any heat exchanger configured to provide heat transfer between air flowing through (or across) the evaporator **216** (i.e., airflow **218** contacting an outer surface of one or more coils of the evaporator **216**) and working fluid passing through the interior of the evaporator **216**. The evaporator **216** may include one or more circuits of coils. The evaporator **216** is fluidically connected to the compressor **206**, such that working fluid generally flows from the evaporator **216** to the condensing unit **204** when the HVAC system **100** is operating to provide cooling.

A portion of the HVAC components **200** are configured to move airflow **218** provided by the blower **228** across the evaporator **216** and out of the duct **222** as conditioned airflow **220**. Return air **224**, which may be air returning from the building, fresh air from outside, or some combination, is pulled into a return duct **226**. A suction side of the blower **228** pulls the return air **224**. The blower **228** discharges airflow **218** into a duct **230** such that airflow **218** crosses the evaporator **216** or heating elements (not shown) to produce conditioned airflow **220**. The blower **228** is any mechanism for providing airflow **218** through the HVAC system **100**. For example, the blower **228** may be a constant speed or variable speed circulation blower or fan. Examples of a variable speed blower include, but are not limited to, belt-

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drive blowers controlled by inverters, direct-drive blowers with electronic commutated motors (ECM), or any other suitable type of blower.

The HVAC components **200** include the air purification subsystem **146** of FIG. **1**. The air purification subsystem **146** includes at least a filter **232** and may also include a reactor unit **234**. The filter **232** is any appropriate air filter capable of removing particulate matter from return air **224**. For example, the filter **232** may be made of fiberglass, cotton, polyester, or any other appropriate material. In some cases, a filter **232** may be rated for the removal of certain sizes of particles from return air **224**. The reactor unit **234** is a device capable of removing microorganisms, VOCs, and/or other contaminants from return air **224**. For example, the reactor unit **234** may include an ultraviolet light source or ozone source that can remove contaminants from the return air **224**.

The HVAC components **200** include the ventilation subsystem **148** of FIG. **1**. The ventilation subsystem **148** includes a damper **240** disposed within a duct **238**. The duct **238** may be exposed to outdoor air or air from another space for dilution of the space conditioned by the HVAC system **100**. When a ventilation action **134** is performed, the damper **240** is opened and the blower **228** (or a separate dedicated blower not shown for conciseness) is turned on to pull a flow of diluting air **236**. In some embodiments, the ventilation subsystem **148** includes a preconditioning unit **242**, which may pre-cool or pre-heat the diluting air **236**, such that the space serviced by the HVAC system **100** is held at a more comfortable temperature.

As described with respect to FIG. **1**, the HVAC system **100** includes one or more sensors **104a-c** in signal communication with the thermostat **106** (e.g., via wired and/or wireless connection). Sensors **104a-c** may be positioned to measure air qualities **114** in the space. In some cases, one or more sensors **104a-c** may be positioned in a duct **222**, **226**, and/or **230** of the HVAC components **200**. The thermostat **106** may only rely on measurements from such in-duct sensor **104a-c** when the blower **228** is on and has been on for a sufficient period of time. The HVAC components **200** may include one or more further sensors (not shown for conciseness), such as sensors for measuring temperature, air humidity, and/or any other properties of a conditioned space (e.g., a room of the conditioned space).

The thermostat **106** may be located within the conditioned space (e.g., a room or building) serviced by the HVAC system **100**. The controller **244** may be separate from or integrated with the thermostat **106**. In addition to the functions described above with respect to FIG. **1**, the thermostat **106** is configured to allow a user to input a desired temperature or setpoint temperature for the conditioned space. In some embodiments, the thermostat **106** includes a user interface and display (e.g., an endpoint **150**) for displaying information related to the operation and/or status of the HVAC system **100**. For example, the user interface may display operational, diagnostic, and/or status messages and provide a visual interface that allows at least one of an installer, a user, a support entity, and a service provider to perform actions with respect to the HVAC system **100**. For example, the user interface may provide for display of messages related to the status and/or operation of the HVAC system **100** (e.g., an air quality score **118**, alert **136**, a mitigation action **130** being executed, and/or the mitigation and air quality history **140**).

The controller **244** is communicatively coupled (e.g., via wired and/or wireless connection) to the HVAC components **200** and configured to control their operation. The controller **244** generally receives the mitigation action **130** determined

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by the thermostat **106** (or an electronic signal indicative of the determined mitigation action **130**) and provides appropriate control signals **252** to other HVAC components **200** to implement the mitigation action **130**. For example, to execute a filtering action **132**, the control signals **252** may cause the blower **228** to turn on to pull return air **224** through the air purification subsystem **146**. To execute a ventilation action **134**, the control signals **252** may cause the damper **240** to at least partially open and the blower **228** to turn on to pull diluting air **236** into the space serviced by the HVAC system **100**.

The controller **244** may include a processor **246**, memory **248**, and input/output (I/O) interface **250**. The processor **246** comprises one or more processors operably coupled to the memory **248**. The processor **246** is any electronic circuitry including, but not limited to, state machines, one or more central processing unit (CPU) chips, logic units, cores (e.g., a multi-core processor), field-programmable gate array (FPGAs), application specific integrated circuits (ASICs), or digital signal processors (DSPs) that communicatively couples to memory **248** and controls the operation of HVAC system **100**. The processor **246** may be a programmable logic device, a microcontroller, a microprocessor, or any suitable combination of the preceding. The processor **246** is communicatively coupled to and in signal communication with the memory **248**. The one or more processors are configured to process data and may be implemented in hardware or software. For example, the processor **246** may be 8-bit, 16-bit, 32-bit, 64-bit or of any other suitable architecture. The processor **246** may include an arithmetic logic unit (ALU) for performing arithmetic and logic operations, processor registers that supply operands to the ALU and store the results of ALU operations, and a control unit that fetches instructions from memory **248** and executes them by directing the coordinated operations of the ALU, registers, and other components. The processor may include other hardware and software that operates to process information, control the HVAC system **100**, and perform any of the functions described herein (e.g., with respect to FIGS. **1-5B**). The processor **246** is not limited to a single processing device and may encompass multiple processing devices.

The memory **248** comprises one or more disks, tape drives, or solid-state drives, and may be used as an over-flow data storage device, to store programs when such programs are selected for execution, and to store instructions and data that are read during program execution. The memory **248** may be volatile or non-volatile and may comprise ROM, RAM, ternary content-addressable memory (TCAM), dynamic random-access memory (DRAM), and static random-access memory (SRAM). The memory **248** is operable to store any suitable set of instructions, logic, rules, and/or code for executing the functions described in this disclosure with respect to FIGS. **1-5B**.

The I/O interface **250** is configured to communicate data and signals with other devices. For example, the I/O interface **250** may be configured to communicate electrical signals with the other components of the HVAC systems **100**. The I/O interface **250** may send control signals **252** to implement mitigation actions **130** as described above. The I/O interface **250** may use any suitable communication protocol. The I/O interface **250** may comprise ports and/or terminals for establishing signal communications between the controller **244** and other devices. The I/O interface **250** may be configured to enable wired and/or wireless communications.

Connections between various components of the HVAC system **100** and between HVAC components **200** may be

wired or wireless. For example, conventional cable and contacts may be used to couple the thermostat **106** to the controller **244** and various components of the HVAC system **100**, including, the compressor **206**, the expansion valve **214**, the blower **228**, damper **240**, and/or sensor(s) **104a-c**. In some embodiments, a wireless connection is employed to provide at least some of the connections between components of the HVAC system **100**. In some embodiments, a data bus couples various components of the HVAC system **100** together such that data is communicated there between. In a typical embodiment, the data bus may include, for example, any combination of hardware, software embedded in a computer readable medium, or encoded logic incorporated in hardware or otherwise stored (e.g., firmware) to couple components of HVAC system **100** to each other.

As an example and not by way of limitation, the data bus may include an Accelerated Graphics Port (AGP) or other graphics bus, a Controller Area Network (CAN) bus, a front-side bus (FSB), a HYPERTRANSPORT (HT) interconnect, an INFINIBAND interconnect, a low-pin-count (LPC) bus, a memory bus, a Micro Channel Architecture (MCA) bus, a Peripheral Component Interconnect (PCI) bus, a PCI-Express (PCI-X) bus, a serial advanced technology attachment (SATA) bus, a Video Electronics Standards Association local (VLB) bus, or any other suitable bus or a combination of two or more of these. In various embodiments, the data bus may include any number, type, or configuration of data buses, where appropriate. In certain embodiments, one or more data buses (which may each include an address bus and a data bus) may couple the controller **244** to other components of the HVAC system **100**.

Example Method of Operation

FIG. **3** is a flowchart of an example method **300** of operating the system of FIG. **1**. Operations of method **300** may be implemented using the processor **108**, memory **110**, and network interface **112** of the thermostat **106**. Method **300** may begin at operation **302** where the thermostat **106** receives measurements of indoor air qualities **114**. At operation **304**, the thermostat **106** determines an indoor air quality score **118**, which may include an individual score **122a-c** for each air quality **114** measured, as described above with respect to FIG. **1**. At operation **306**, the air quality score **118** may be provided for display, for example, on an endpoint device **150**, as described with respect to FIG. **1** above.

At operation **308**, the thermostat **106** determines whether the scores **122a-c** from operation **304** indicate that a mitigation action **130** is indicated. For example, if at least one of the scores **122a-c** is “poor”, a mitigation action **130** may be indicated. If no mitigation action **130** is indicated, the thermostat **106** returns to the start of method **300** and continues receiving measurements of air qualities **114**. If the scores **122a-c** indicate that a mitigation action **130** is indicated, the thermostat **106** proceeds to operation **310**.

At operation **310**, the thermostat **106** determines the mitigation action **130** to improve air quality. For example, a mitigation action **130** may be determined that is able to improve any air qualities **114** with a score **122a-c** of “poor”. For example, if scores **122a-c** of “poor” are determined for air qualities **114** of PM and/or VOCs, a filtering action **132** may be identified. A ventilation action **134** may be identified to decrease the concentration of CO₂ or if filtering was not effective at removing another contaminant. When possible, a filtering action **132** is typically performed before a ventilation action **134**. TABLE **2** above summarizes when example mitigation actions **130** may be determined, and

further examples of the determination of a mitigation action **130** are provided below with respect to FIGS. **4**, **5A**, and **5B**.

At operation **312**, the determined mitigation action **130** is executed. For example, a filtering action **132** may be executed by turning on the blower **228** of the HVAC system **100** in order to cause air to pass through the air purification subsystem **146**. If a ventilation action **134** is determined to be needed, the ventilation action **134** may be executed by, for example, opening a damper **240** of the ventilation subsystem **148** to allow a diluting flow of diluting air **236** to enter the space.

At operation **314**, further measurements of air qualities **114** are received. For example, the measurements of air qualities **114** may be received while the mitigation action **130** is occurring (e.g., during a cycle of the mitigation action **130**) and/or at or near the end of the mitigation action **130**.

At operation **316**, the thermostat **106** determines whether criteria **152** are satisfied for continuing the mitigation action **130**. For example, the mitigation action **130** may be continued if an air quality **114** still has a score **122a-c** of “poor” and if the air quality **114** is improving as a result of the mitigation action **130** executed at operation **312**. Other example criteria **152** for continuing or repeating a mitigation action **130** are described further below with respect to FIGS. **4**, **5A**, and **5B** and TABLE **3**. If the criteria **152** are satisfied, the thermostat **106** may return to operation **312** and execute the mitigation action **130** again (e.g., execute another cycle of the mitigation action **130**) or continue executing the mitigation action **130**. Otherwise, the thermostat **106** proceeds to operation **318** and stops or pauses the mitigation action **130**.

At operation **320**, the thermostat **106** determines whether criteria **152** are satisfied for starting a secondary mitigation action **130**. For example, if the mitigation action **130** executed at operation **312** was a filtering action **132** and was stopped or paused at operation **318**, the thermostat **106** may determine whether criteria **152** are satisfied for a ventilation action **134** to be performed. For example, if the outdoor air quality **128** is better than “poor”, a ventilation action **134** may be allowed. Other example criteria **152** for allowing a ventilation action **134** are described further below with respect to FIGS. **4**, **5A**, and **5B** and TABLE **3**. If the criteria **152** are satisfied at operation **320**, the thermostat proceeds to operation **322**. Otherwise, the thermostat **106** proceeds to operation **326**.

At operation **322**, the thermostat **106** executes the secondary mitigation action **130** (e.g., the ventilation action **134** in this example). The ventilation action **134** may be executed by at least partially opening the damper **240**, as described above with respect to FIGS. **1** and **2**. At operation **324**, the thermostat **106** determines whether to continue or repeat the secondary mitigation action **130** (see operation **316**). For example, if the air qualities **114** are improving after the initial run of the mitigation action **130**, then the criteria **152** may be satisfied. Other example criteria **152** for continuing or repeating a mitigation action **130** are described further below with respect to FIGS. **4**, **5A**, and **5B** and TABLE **3**.

At operation **326**, the thermostat **106** evaluates the mitigation and air quality history **140** to determine whether mitigation actions **130** have been ineffective. For example, if mitigation actions **130** are performed repeatedly without a sufficient improvement in air qualities **114**, then the mitigation actions **130** may be determined to be ineffective. If this is the case, the thermostat **106** proceeds to operation **328** and provides a notification (e.g., an alert **136** of FIGS. **1** and **2**) indicating that the space has poor air quality and that some system change may be needed to improve the air quality.

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Modifications, additions, or omissions may be made to method **300** depicted in FIG. 3. Method **300** may include more, fewer, or other operations. For example, operations may be performed in parallel or in any suitable order. While at times discussed as the thermostat **106** performing the operations, any suitable components (e.g., a controller **244** of FIG. 2) of the HVAC system **100** may perform one or more operations of the method **300**.

Example Cycle-Based Method of Operation

In some cases, the HVAC system **100** is operated in a cycle-based fashion, such that mitigation actions **130** may be performed for a predefined time period or cycle (e.g., of 15 minutes or the like) and subsequent mitigation actions **130** (if any) are re-evaluated following or near the completion of each cycle. FIG. 4 shows a state diagram **400** of different example transitions between operating states of the HVAC system **100** for air quality mitigation. In an example, the HVAC system **100** may initially be in an idle state **402** in which no mitigation action **130** is being performed. From the idle state, the thermostat **106** determines whether to enter a “cleaning run” (CR) or “ventilation run” (VR) status. A CR status corresponds to a determination to perform a filtering action **132**, and a VR status corresponds to a determination to perform a ventilation action **134**.

If there is a CR status, the HVAC system **100** transitions to state **404** and one filtering action **132** cycle is performed. The filtering action **132** may be performed for a time T_c (see TABLE 3 below), during which the blower **228** is run at a predefined airflow rate CFMc (see TABLE 3). After the cycle is completed, the air qualities **114** are re-evaluated (e.g., by performing air quality scoring **116**), and the thermostat **106** determines whether to run another cycle of the filtering action **132** (e.g., a CR), pause cleaning by filtration for a predefined period to be re-attempted later (e.g., a cleaning pause or “CP” of FIG. 4), switch to cleaning by ventilation (e.g., a VR), or stop cleaning by filtration (e.g., a cleaning stop or “CS” in FIG. 4).

As an example, a status of CR may be determined if each of the following criteria **152** are satisfied: (1) an active contaminant is present; (2) filtering has not been paused for the active contaminant due to previous unsuccessful filtration attempts; (3) active contaminant cannot be removed by ventilation due to any restrictions on cleaning by ventilation; and (4) either (a) a total number of consecutive filtering cycles has not exceeded Nc_x , defined for the active of the contaminant x, such as $PM_{2.5}$, TVOC, CO_2 , (see TABLE 3) or (b) the active contaminant is trending down. As shown in FIG. 4, when a CR is identified, the HVAC system **100** changes to (or remains in) state **404**.

As another example, a status of CP may be determined if all of the following criteria **152** apply: (1) the active contaminant is either TVOC or CO_2 ; (2) the total number of filtration cycles exceeds Nc_x , defined for the active contaminant x, such as $PM_{2.5}$, TVOC, CO_2 (see TABLE 3); and (3) the active contaminant is not trending down. As shown in FIG. 4, when a status of CP is determined, the HVAC system **100** is brought into state **406** and cleaning by filtration is stopped. Cleaning may be stopped for at least Pc_x minutes, defined for the active contaminant x, such as $PM_{2.5}$, TVOC, CO_2 (see TABLE 3).

As another example, a status of CS may be determined if an active contaminant is no longer present (e.g., if the air quality scores **118** indicate no mitigation action **130** is needed). As shown in FIG. 4, when a status of CS is determined, the HVAC system **100** is brought into state **408**, and the HVAC system **100** may then return to the idle state **402**.

If there is a VR status, the HVAC system **100** transitions to state **412** and one cycle of a ventilation action **134** is performed. The ventilation cycle may last T_v minutes (see TABLE 3). After each ventilation cycle, the air qualities **114** and air quality score **118** are re-evaluated to determine

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whether to run another ventilation cycle (e.g., a VR); pause ventilation for a predefined period (e.g., a ventilation pause or “VP” of FIG. 4) to be reattempted later, temporarily ban ventilation for a predefined period (e.g., a ventilation ban or “VB” of FIG. 4) to be reattempted later, switch to cleaning by filtration (e.g., a CR), or stop cleaning by ventilation (e.g., ventilation stop or “VS” of FIG. 4).

As an example, a VR status may be determined if all of the following criteria **152** apply: (1) an active contaminant is present; (2) a ventilation subsystem **148** is available and is operated by the thermostat **106** and/or controller **244**; (3) the user has not opted out of ventilation (e.g., via user override **138** of FIG. 1); (4) cleaning by ventilation has not been paused for the active contaminant due to previous unsuccessful attempts; (5) cleaning by ventilation has not been banned due to poor outside air quality; (6) the outside air quality **128** (if available) is not worse than “Fair”; (7) the active contaminant cannot be removed by a filtering action **132** because of either (a) the absence of a designated air purification subsystem **146** or (b) exceeding a maximum allowed number of consecutive filtration cycles (see Nc_{pm} , Nc_{voc} and Nc_{co2} of TABLE 3) without improvement in the active contaminant trend; and (8) the total number of consecutive ventilation cycles has not exceeded a predefined value (see Nv_{pm} , Nv_{voc} and Nv_{co2} of TABLE 3). As shown in FIG. 4, when there is a status of VR, the HVAC system **100** moves to (or remains in) state **412**.

As another example, a VB status may be determined when the HVAC system **100** should at least temporarily ban ventilation actions **134**. A VB status may be determined if the concentrations of PM and/or VOCs increased (e.g., trended upwards) during a previous ventilation cycle. If a VB status is determined, the HVAC system **100** may be moved to state **416**. Ventilation may be banned for B_v minutes (see TABLE 3).

As another example, a VP status may be determined if both: (1) an active contaminant is present and (2) the total number of ventilation cycles exceeded Nv_x defined for the active contaminant x, such as $PM_{2.5}$, TVOC or CO_2 (see TABLE 3). If a VP status is determined, the HVAC system moves to state **414** of FIG. 4. Ventilation may be paused for at least Pv_x minutes defined for the active contaminant x, such as $PM_{2.5}$, TVOC or CO_2 (see TABLE 3). A VS status may be determined when an active contaminant is no longer present. When a VS status is determined, the HVAC system **100** is moved to state **418** and subsequently back to idle state **402**.

In some cases, a user (e.g., a building administrator or an occupant of the space serviced by the HVAC system **100**) may be given the option to postpone a mitigation action **130**. In such cases, a user override **138** of FIG. 1 may be provided to postpone mitigation and bring the HVAC system **100** to state **410**. For example, a current filtering or ventilation cycle of state **404** or **412**, respectively, may be postponed for a selected period of time. After this period of time, mitigation may be resumed based on the air qualities **114** at that time.

Certain decisions described above for transitioning between the various states illustrated in FIG. 4 involve evaluation of a trend in an air quality (e.g., a contaminant or measurable air component) over time. These trends may be a score **122a-c** indicating whether a level of a given contaminant is increasing, decreasing, or remains about the same. As an example, a trend score tQ_x for a component x being one of $PM_{2.5}$, VOC, or CO_2 may be determined as a coefficient using linear regression fitting all available short-term moving averages (STA_x) for that component x over the previous T_{tnd} minutes when at least M % of STA_x values are available (see TABLES 1 and 3). T_{tnd} may be either: (1) T_v when ventilation is banned (state **416** of FIG. 4) or (2) $2 \times T_c$ for other cases. Example values of T_v and T_c are provided in TABLE 3.

TABLE 3 below shows example parameters, which are described both above with respect to FIG. 4 and below with respect to FIGS. 5A and 5B for performing mitigation cycles to help remove or reduce the amount of the example

contaminants of PM, VOCs, and CO₂. This disclosure contemplates that similar parameters may be used for the mitigation of other contaminants using the same or similar processes.

TABLE 3

Example Mitigation Configuration Parameters					
Parameter	Purpose	Units	Min	Max	Default
Tc	run time interval for one filtering cycle	min	15	60	15
Tv	run time interval for one ventilation cycle	min	15	45	15
CFMc	airflow rate required for cleaning by filtration	CFM	CFM min (Minimum airflow rate for the given HVAC system)	CFM max (Maximum airflow rate for the given HVAC system)	CFM min + (CFM_max - CFM_min)/3
Nc_pm	maximum number of consecutive PM2.5 filtering cycles before mitigation is reevaluated	#	5	50	10
Nc_voc	maximum number of consecutive VOC filtering cycles	#	3	50	7 without dedicated ventilation subsystem 15 with dedicated air ventilation subsystem
Nc_co2	maximum number of consecutive CO2 filtering cycles	#	1	50	2
Nv_pm	max number of consecutive PM2.5 ventilation cycles	#	3	50	7
Nv_voc	max number of consecutive VOC ventilation cycles	#	3	50	10
Nv_co2	max number of consecutive CO ₂ ventilation cycles	#	2	50	10
Pc_voc	minimum time interval for which filtering should be paused before being re-attempted	Min	60	300	120
Pc_co2	min time interval for which Co2 filtering should be paused before being re-attempted	Min	60	300	120
Pv_pm	min time interval for which PM2.5 ventilation should be paused before being re-attempted	Min	60	300	180
Pv_voc	min time interval for which VOC ventilation should be paused before being re-attempted	Min	60	300	180
Pv_co2	min time interval for which CO2 ventilation should be paused before being re-attempted	Min	60	300	180
Bv	min time interval for which ventilation should be temporarily banned due to poor outdoor air quality detected	min	30	300	60
ITC	max value for improving trend coefficients	#	-0.1	0	-0.05
WTC	min value for worsening trend coefficients	#	0	0.1	0.05
M	min % of values available to compute their trending score	%	30	100	60
Np	max number of consecutive occurrences LTA score falls in poor range before an alert is raised	#	1	30	10

FIGS. 5A and 5B illustrate processes 500a,b for operating the HVAC system 100 using cycle-based air quality mitigation. The example processes 500a,b are performed to mitigate the contaminants of PM (specifically PM2.5 in the examples of FIGS. 5A and 51B), VOCs, and CO₂. FIG. 5A illustrates an initial process 500a for operating the HVAC system 100. Process 500a begins at operation 502 where measurements of indoor air qualities 114 (referred to as “IAQ” in FIGS. 5A and 5B) are received by the thermostat 106. The air qualities 114 are concentrations of PM2.5, VOCs, and CO₂. At operation 504, the measurements of one or more of these air qualities 114 may be displayed on an endpoint device 150, as described with respect to FIG. 1 above.

At operation 506, moving averages 120a-c and corresponding scores 122a-c are determined for the different monitored contaminants. At operation 508, the thermostat 106 determines whether a filtering action is already in progress. If this is the case, the thermostat 106 proceeds to operation 510. Otherwise, the thermostat 106 moves to operation 518.

At operation 510, the thermostat 106 determines whether the filtering time has exceeded a maximum value Tc (see TABLE 3 above). If this is not the case, the thermostat 106 allows the filtering action 132 to continue. Otherwise, the thermostat 106 proceeds to stop the filtering action 132 at operation 514. At operation 516, the thermostat 106 increases a count of the number of filtering actions 132 performed for the active contaminant.

At operation 516, the thermostat 106 determines whether a ventilation action 134 is in progress. If this is the case, thermostat 106 proceeds to operation 520. Otherwise, the thermostat 106 moves to operation 526. At operation 520, the thermostat 106 determines whether the ventilation time has exceeded a maximum value Tv (see TABLE 3 above). If this is not the case, the thermostat 106 proceeds to operation 532. Otherwise, the thermostat 106 proceeds to stop the ventilation action 134 at operation 522. At operation 524, the thermostat 106 increases a count of the number of ventilation actions 134 performed for the active contaminant.

At operation 526, the thermostat 106 determines whether there is a ventilation ban in progress (see, e.g., state 416 of FIG. 4). If this is not the case, the thermostat proceeds to operation 538. Otherwise, the thermostat 106 proceeds to operation 528 and determines whether the ventilation ban has exceeded a ban time limit By (see TABLE 3 above). If this time limit is exceeded, the thermostat 106 proceeds to operation 530 and removes the ventilation ban.

At operation 532, the thermostat 106 determines whether there is an upward trend over time in the concentration of PM2.5 and/or VOC. If this is not the case, the current ventilation action 134 is allowed to continue at operation 534. However, if this is the case, a ventilation ban is activated at operation 536. At operation 538, air qualities 114 are evaluated (e.g., as described for operation 506), and a mitigation action 130 is determined as described above with respect to FIG. 1.

FIG. 5B shows a subsequent process 500b for implementing the mitigation action 130 determined at operation 538 based on the type of active contaminant. Process 500b is linked to operations of process 500a as illustrated in FIG. 5B. At operation 546 of process 500b, the thermostat 106 determines if the active contaminant is PM2.5. If this is the case, the thermostat 106 proceeds to operation 548. Otherwise, the thermostat 106 proceeds to operation 556.

At operation 548, the thermostat 106 determines whether the number of filtering cycles for the active contaminant (abbreviated as “a.c.” in FIG. 5B) exceeds a threshold value (see Nc_pm, Nc_voc, Nc_co2 of TABLE 3 above). If the threshold is not exceeded, the thermostat 106 proceeds to operation 550. If the threshold is exceeded, the thermostat 106 proceeds to operation 560.

At operation 550, the thermostat 106 determines whether filtering is paused. If filtering is not paused, the thermostat 106 continues the filtering action 132 at operation 512 of process 500a. However, if filtering is paused, the thermostat 106 determines whether filtering has been paused for greater than a threshold time (see Pc_pm, Pc_voc, Pc_co2 of TABLE 3 above). If this is not the case, the thermostat 106 returns to the start of process 500a. However, if this is the case, the thermostat 106 runs a cycle of the filtering action 132 at operation 554 before returning to process 500a.

At operation 560 (if the maximum number of filtering cycles is exceeded at operation 548), the thermostat 106 determines whether there is a downward trend in the concentration of the active contaminant PM2.5. If this is the case, the thermostat 106 proceeds to operation 550 described above. If this is not the case, the thermostat 106 proceeds to operation 564 and determines whether ventilation is available (e.g., whether the HVAC system 100 includes a ventilation subsystem 148—see FIGS. 1 and 2).

At operation 556 (if PM2.5 is not the active contaminant at operation 546), the thermostat 106 determines whether VOC is the active contaminant. If VOC is the active contaminant, the thermostat 106 proceeds to operation 558 and determines whether air purification is available. Air purification is determined to be available if the HVAC system 100 includes an air purification subsystem 146 with a reactor unit 234 capable of removing VOCs (see FIGS. 1 and 2 and their corresponding descriptions above). If air purification is available, the thermostat 106 proceeds to operation 548 described above. If air purification is not available, the thermostat 106 proceeds to operation 564 and determines whether ventilation is available.

At operation 526 (if VOC is not the active contaminant at operation 556), the thermostat 106 determines whether the active contaminant is CO₂. If CO₂ is also not the active contaminant, the thermostat 106 determines no cleaning is needed at operation 582 and returns to normal operation and process 500a. However, if CO₂ is the active contaminant, the thermostat 106 proceeds to operation 564 and determines whether ventilation is available. If ventilation is not available, the thermostat 106 proceeds to operation 584 described below. If ventilation is available, the thermostat 106 proceeds to operation 566 and determines whether ventilation is banned. If ventilation is banned, the thermostat 106 proceeds to operation 584. If ventilation is not banned, the thermostat 106 proceeds to operation 568 and determines whether an outdoor air quality 128 is available. If the outdoor air quality 128 is not available, the thermostat 106 proceeds to operation 572. However, if the outdoor air quality 128 is available, the thermostat 106 proceeds to operation 570 and determines whether the outdoor air is sufficiently clean based on the outdoor air quality 128 (e.g., whether a score for the outdoor air is better than “poor”). If the outdoor air is not sufficiently clean, the thermostat 106 proceeds to operation 584. Otherwise, the thermostat 106 proceeds to operation 572.

At operation 572, the thermostat 106 determines whether the number of ventilation cycles performed exceeds a threshold (see Nv_pm, Nv_voc, Nv_co2 of TABLE 3 above). If the threshold is not exceeded, the thermostat 106 proceeds to operation 574 and determines whether ventila-

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tion is paused. If ventilation is not paused, the thermostat **106** proceeds to operation **578** and runs a cycle of a ventilation action **134**. If ventilation is paused, the thermostat **106** proceeds to operation **576** and determines whether a maximum ventilation pause time for the active contaminant (see Pv_pm, Pv_voc, Pv_co2 of TABLE 3 above) has been exceeded. If the maximum ventilation pause time is not reached, the thermostat **106** proceeds to operation **584**. Otherwise, the thermostat **106** runs a cycle of the ventilation action **134** at operation **578**.

At operation **584** (if ventilation is not available at operation **564**, ventilation is banned at operation **566**, outdoor air is not clean at operation **570**, ventilation is paused at operation **580**, or the maximum ventilation pause time has not been reached at operation **576**), the thermostat **106** determines if the active contaminant is PM2.5. If this is the case, the thermostat **106** proceeds to operation **586** and determines if the score **122a-c** based on the long-term average (LTA for the active contaminant) has been poor for longer than a threshold number of cycles Np (see TABLE 3 above). If this is not the case, the thermostat **106** proceeds to operation **596**. Otherwise, the thermostat **106** proceeds to operation **588** and sets or provides an alert **136** indicating a persistently high concentration of PM2.5 in the indoor space that could not be addressed by the previous cleaning attempts before moving to operation **550**, described above.

If the active contaminant is not PM2.5 at operation **584**, the thermostat **106** proceeds to operation **590** and reset the ventilation cycle counter for the active contaminant. At operation **592**, the thermostat **106** then determines if the number of filtering cycles performed exceeds a threshold value (see operation **548**). If this is the case, the thermostat **106** proceeds to operation **594** and determines if there is a downwards trend in the concentration of the active contaminant (see operation **560**). If there is a downward trend, the thermostat **106** proceeds to operation **550**. Otherwise, the thermostat **106** proceeds to operation **596**. If the number of filtering cycles is not greater than the threshold at operation **592**, the thermostat **106** also proceeds to operation **596**.

At operation **596**, the thermostat **106** determines whether filtering is paused (see operation **550**). If filtering is paused, the thermostat **106** proceeds to operation **552**. Otherwise, the thermostat **106** proceeds to operation **597**. At operation **597**, the thermostat **106** pauses the filtering action **132**. At operation **598**, the thermostat **106** determines if the score **122a-c** based on the long-term average (LTA for the active contaminant) has been poor for longer than a threshold number of cycles Np (see TABLE 3 above). If this is not the case, the thermostat **106** returns to operation **596**. Otherwise, the thermostat **106** proceeds to operation **599** and sets or provides an alert **136** for persistently poor indoor air quality.

Modifications, additions, or omissions may be made to processes **500a,b** depicted in FIGS. 5A and 5B. Processes **500a,b** may include more, fewer, or other operations. For example, operations may be performed in parallel or in any suitable order. While at times discussed as the thermostat **106** performing the operations, any suitable components (e.g., a controller **244** of FIG. 2) of the HVAC system **100** may perform one or more operations of the processes **500a,b**.

While several embodiments have been provided in the present disclosure, it should be understood that the disclosed systems and methods might be embodied in many other specific forms without departing from the spirit or scope of the present disclosure. The present examples are to be considered as illustrative and not restrictive, and the intention is not to be limited to the details given herein. For

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example, the various elements or components may be combined or integrated in another system or certain features may be omitted, or not implemented.

In addition, techniques, systems, subsystems, and methods described and illustrated in the various embodiments as discrete or separate may be combined or integrated with other systems, modules, techniques, or methods without departing from the scope of the present disclosure. Other items shown or discussed as coupled or directly coupled or communicating with each other may be indirectly coupled or communicating through some interface, device, or intermediate component whether electrically, mechanically, or otherwise. Other examples of changes, substitutions, and alterations are ascertainable by one skilled in the art and could be made without departing from the spirit and scope disclosed herein.

To aid the Patent Office, and any readers of any patent issued on this application in interpreting the claims appended hereto, applicants note that they do not intend any of the appended claims to invoke 35 U.S.C. § 212(f) as it exists on the date of filing hereof unless the words “means for” or “step for” are explicitly used in the particular claim.

What is claimed is:

1. A heating, ventilation, and air conditioning (HVAC) system configured to provide conditioned air to a space, the HVAC system comprising:

one or more air quality sensors, each air quality sensor configured to measure an air quality; and

a thermostat comprising:

an interface communicatively coupled to the one or more air quality sensors; and

a processor configured to:

receive indoor air quality measurements from the one or more air quality sensors;

determine an indoor air quality score based at least in part on the received indoor air quality measurements;

determine, based at least in part on the indoor air quality score, a mitigation action, wherein the mitigation action comprises: (i) a filtering action comprising filtering air provided to the space using an air purification subsystem, and/or (ii) a ventilation action comprising ventilating the space using a ventilation subsystem;

determine that the mitigation action comprises at least the filtering action by determining that a current indoor air quality is below a quality threshold and that a number of previous attempts to perform the filtering action do not exceed a maximum threshold; and

execute the filtering action by performing a filtering cycle using the air purification subsystem.

2. The HVAC system of claim 1, wherein the processor is further configured to, after completion of the filtering cycle:

if the indoor air quality measurements received after the filtering cycle is complete satisfy second criteria associated with one or more of a type of contaminant needing to be removed, a number of previous attempts to perform the filtering action, and a trend in the indoor air quality over time, pause execution of the filtering action and determine whether a ventilation cycle is allowed; and

if the indoor air quality measurements received after the filtering cycle is complete indicates mitigation is no longer needed, stop the filtering action.

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3. The HVAC system of claim 2, wherein the processor is further configured to:

determine that the ventilation cycle is allowed by determining whether third criteria are satisfied associated with one or more of an outdoor air quality, a number of previous attempts to perform the ventilation action, and a trend in the indoor air quality over time; and

after determining that the ventilation cycle is allowed, execute the ventilation cycle by opening a damper of the ventilation subsystem of the HVAC system to allow a diluting flow of air to enter the space.

4. The HVAC system of claim 1, wherein the processor is further configured to:

determine that an outdoor air quality is below a quality threshold; and

in response to determining that the outdoor air quality is below the quality threshold, prevent execution of the ventilation action.

5. The HVAC system of claim 1, wherein the processor is further configured to:

execute the filtering action by turning on a blower that provides a flow of the air across the air purification subsystem; and

execute the ventilation action by opening a damper in the ventilation subsystem of the HVAC system to allow a diluting flow of air to enter the space.

6. The HVAC system of claim 1, wherein the processor is further configured to determine the indoor air quality score by:

determining a moving average of the indoor air quality over time; and

determining the indoor air quality score by comparing the moving average to a predefined threshold.

7. The HVAC system of claim 1, wherein the processor is further configured to:

determine that the one or more air quality sensors are located within a duct of the HVAC system;

determine that an indoor air quality measurement is needed; and

activate a blower of the HVAC system to provide a flow of air past the one or more air quality sensors located within the duct prior to measuring the indoor air quality.

8. The HVAC system of claim 1, wherein the indoor air quality comprises one or more of a concentration of particles in air, a concentration of volatile organic compounds in air, or a concentration of carbon dioxide in air.

9. A method of operating a heating, ventilation, and air conditioning (HVAC) system configured to provide conditioned air to a space, the method comprising:

measuring an air quality using one or more air quality sensors; and

receiving indoor air quality measurements from the one or more air quality sensors located within the space;

determining an indoor air quality score based at least in part on the received indoor air quality measurements;

determining, based at least in part on the indoor air quality score, a mitigation action, wherein the mitigation action comprises: (i) a filtering action comprising filtering air provided to the space using an air purification subsystem, and/or (ii) a ventilation action comprising ventilating the space using a ventilation subsystem;

determining that the mitigation action comprises at least the filtering action by determining that a current indoor air quality is below a quality threshold and that a number of previous attempts to perform the filtering action do not exceed a maximum threshold; and

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executing the filtering action by performing a filtering cycle using the air purification subsystem.

10. The method of claim 9, further comprising, after completion of the filtering cycle:

if the indoor air quality measurements received after the filtering cycle is complete satisfy second criteria associated with one or more of a type of contaminant needing to be removed, a number of previous attempts to perform the filtering action, and a trend in the indoor air quality over time, pausing execution of the filtering action and determining whether a ventilation cycle is allowed; and

if the indoor air quality measurements received after the filtering cycle is complete indicates mitigation is no longer needed, stopping the filtering action.

11. The method of claim 10, further comprising determining that the ventilation cycle is allowed by determining whether third criteria are satisfied associated with one or more of an outdoor air quality, a number of previous attempts to perform the ventilation action, and a trend in the indoor air quality over time; and

after determining that the ventilation cycle is allowed, executing the ventilation cycle by opening a damper of a ventilation subsystem of the HVAC system to allow a diluting flow of air to enter the space.

12. The method of claim 9, further comprising: determining that an outdoor air quality is below a quality threshold; and

in response to determining that the outdoor air quality is below the quality threshold, preventing execution of the ventilation action.

13. The method of claim 9, further comprising: executing the filtering action by turning on a blower that provides a flow of the air across the air purification subsystem; and

executing the ventilation action by opening a damper in the ventilation subsystem of the HVAC system to allow a diluting flow of air to enter the space.

14. The HVAC method of claim 9, further comprising determining the indoor air quality score by:

determining a moving average of the indoor air quality over time; and

determining the indoor air quality score by comparing the moving average to a predefined threshold.

15. The method of claim 9, further comprising: determining that the one or more air quality sensors are located within a duct of the HVAC system;

determining that an indoor air quality measurement is needed; and

activating a blower of the HVAC system to provide a flow of air past the one or more air quality sensors located within the duct prior to measuring the indoor air quality.

16. The method of claim 9, wherein the indoor air quality comprises one or more of a concentration of particles in air, a concentration of volatile organic compounds in air, or a concentration of carbon dioxide in air.

17. A thermostat of a heating, ventilation, and air conditioning (HVAC) system configured to provide conditioned air to a space, the thermostat comprising:

an interface communicatively coupled to one or more air quality sensors, each air quality sensor configured to measure an air quality; and

a processor configured to: receive indoor air quality measurements from the one or more air quality sensors;

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determine an indoor air quality score based at least in part on the received indoor air quality measurements; determine, based at least in part on the indoor air quality score, a mitigation action, wherein the mitigation action comprises: (i) a filtering action comprising filtering air provided to the space using an air purification subsystem, and/or (ii) a ventilation action comprising ventilating the space using a ventilation subsystem; 5

determine that the mitigation action comprises at least the filtering action by determining that a current indoor air quality is below a quality threshold and that a number of previous attempts to perform the filtering action do not exceed a maximum threshold; 10

and 15

execute the filtering action by performing a filtering cycle using the air purification subsystem.

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